Method, System, Readable Storage Medium and Electronic Equipment for Detecting Crowd Panic Scattering

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Abstract

The present invention discloses a method for detecting crowd panic scattering, which includes the following steps: performing moving target detection on a real-time video stream to obtain the motion trajectory of each pedestrian within the crowd; calculating the velocity factor of the crowd; calculating the vector direction information entropy of the crowd; calculating a crowd confusion degree evaluation index; and when the crowd confusion degree evaluation index is greater than a preset threshold, issuing an abnormal warning for crowd panic scattering. By using moving target detection to obtain pedestrian motion trajectories, and subsequently obtaining the crowd velocity factor and vector direction information entropy, the present invention derives the crowd confusion degree evaluation index. Furthermore, utilizing the characteristic that the crowd confusion degree evaluation index increases during crowd panic scattering, it automatically issues warnings. This saves the labor costs of manual monitoring, and improves the accuracy, efficiency, and sensitivity of detection. The present invention also discloses a crowd panic scattering detection system, a readable storage medium, and an electronic device.

Claims

1. A method for detecting crowd panic scattering, characterized in that it comprises the following steps:

S1, acquiring a real-time video stream of a monitoring scene, and performing moving target detection on the real-time video stream to obtain the motion trajectory of each pedestrian in the crowd;

S2, according to the motion trajectory of each pedestrian in the crowd, calculating a crowd velocity factor V_t;

S3, according to the motion trajectory of each pedestrian in the crowd, calculating a crowd vector direction information entropy W_t;

S4, according to the crowd velocity factor V_t and the crowd vector direction information entropy W_t, calculating a crowd confusion degree evaluation index L_t;

S5, when the crowd confusion degree evaluation index L_t is greater than a preset threshold, issuing an abnormal warning for crowd panic scattering.

2. The method for detecting crowd panic scattering according to claim 1, characterized in that the step S2 comprises:

S21, calculating the velocity v_j of each pedestrian according to the motion trajectory of each pedestrian in the crowd;

S22, according to the velocity of each pedestrian, calculating an overall average velocity $\bar{v_t} = \Sigma_{j=1}^{N} v_j$, where N represents the number of pedestrians in the crowd;

S23, according to the overall average velocity, calculating the crowd velocity factor V_t.

3. The method for detecting crowd panic scattering according to claim 2, characterized in that the motion trajectory of each pedestrian comprises a position sequence for each pedestrian, and the step S21 comprises:

According to the position sequence of each pedestrian, calculating the velocity of each pedestrian $v_j = \partial Q_j / \partial t$, wherein Q_j represents the position of the j-th pedestrian, obtained according to the position sequence of each pedestrian.

4. The method for detecting crowd panic scattering according to claim 2, characterized in that the step S23 comprises:

$$V_t = \partial \bar{v_t} / \partial t$$

5. The method for detecting crowd panic scattering according to claim 1, characterized in that the step S3 comprises:

S31, determining a vector direction of each pedestrian according to the motion trajectory of each pedestrian in the crowd, wherein the vector direction corresponds to 8 continuous regions spaced at 45-degree intervals within a 360-degree circumferential range;

S32, according to the vector direction of each pedestrian, counting the proportion P_j of the number of pedestrians in each vector direction;

S33, according to the proportion P_j of the number of pedestrians in each vector direction, calculating the crowd vector direction information entropy W_t,

$$W_t = -\Sigma_{j=1}^{8} P_j \cdot \log P_j$$

6. The method for detecting crowd panic scattering according to claim 1, characterized in that the step S4 comprises:

$L_t = aW_t + (1 - a)V_t$, wherein a represents an adjustment parameter used to adjust the impact proportion of the crowd velocity factor V_t and the crowd vector direction information entropy on the crowd confusion degree evaluation index L_t.

7. The method for detecting crowd panic scattering according to claim 6, characterized in that the adjustment parameter is 0.5.

8. A system for detecting crowd panic scattering, characterized in that it comprises:

A **motion trajectory acquisition module**, configured to acquire a real-time video stream of a monitoring scene, and perform moving target detection on the real-time video stream to obtain the motion trajectory of each pedestrian in the crowd;

A **velocity factor calculation module**, configured to calculate a crowd velocity factor V_t according to the motion trajectory of each pedestrian in the crowd, where t represents the current moment, corresponding to the current frame of the real-time video stream;

An **information entropy calculation module**, configured to calculate a crowd vector direction information entropy W_t according to the motion trajectory of each pedestrian in the crowd;

A **confusion degree evaluation index calculation module**, configured to calculate a crowd confusion degree evaluation index L_t according to the crowd velocity factor V_t and the crowd vector direction information entropy W_t;

An **abnormal warning module**, configured to issue an abnormal warning for crowd panic scattering when the crowd confusion degree evaluation index L_t is greater than a preset threshold S.

9. A readable storage medium, comprising instructions, characterized in that when the instructions run on an electronic device, they cause the electronic device to execute the method for detecting crowd panic scattering according to any one of claims 1-7.

10. An electronic device, comprising a memory, a processor, and a computer program stored on the memory and executable on the processor, characterized in that when the processor executes the program, it implements the method for detecting crowd panic scattering according to any one of claims 1-7.

Description

Technical Field

The present invention relates to the technical field of security monitoring, and more specifically, relates to a method, system, readable storage medium, and electronic device for detecting crowd panic scattering.

Background Art

With the rapid increase in urban population density, many public infrastructures frequently experience short-term peaks in pedestrian flow. Global abnormal behaviors that may exist in group movements (such as

abnormal crowd panic and scattering evacuation caused by emergencies) pose major safety hazards. Therefore, it is highly necessary to detect crowd abnormal panic scattering in public infrastructures and other places to carry out targeted subsequent disposal and response.

Traditional crowd panic scattering detection often adopts manual monitoring methods, which are time-consuming and labor-intensive, and cannot guarantee detection accuracy, efficiency, and sensitivity.

Summary of the Invention

Based on the above problems, the present invention proposes a method, system, readable storage medium, and electronic device for detecting crowd panic scattering.

The technical solution adopted by the present invention to solve the above technical problems is: A method for detecting crowd panic scattering, comprising the following steps:

- **S1**, acquiring a real-time video stream of a monitoring scene, and performing moving target detection on the real-time video stream to obtain the motion trajectory of each pedestrian in the crowd;
- **S2**, calculating a crowd velocity factor $\boldsymbol{V_t}$ according to the motion trajectory of each pedestrian in the crowd;
- **S3**, calculating a crowd vector direction information entropy $\boldsymbol{W_t}$ according to the motion trajectory of each pedestrian in the crowd;
- **S4**, calculating a crowd confusion degree evaluation index $\boldsymbol{L_t}$ according to the crowd velocity factor $\boldsymbol{V_t}$ and the crowd vector direction information entropy $\boldsymbol{W_t}$;
- **S5**, when the crowd confusion degree evaluation index $\boldsymbol{L_t}$ is greater than a preset threshold $\boldsymbol{S}$, issuing an abnormal warning for crowd panic scattering.

Beneficial Effects: By acquiring real-time video streams and performing moving target detection to obtain pedestrian motion trajectories, acquiring the crowd velocity factor and vector direction information entropy based on the pedestrian motion trajectories, and then obtaining the crowd confusion degree evaluation index, while utilizing the characteristic that the crowd confusion degree evaluation index increases when a crowd panics and scatters, the present invention automatically issues abnormal warnings for crowd panic scattering. This saves the labor costs of manually monitoring crowd panic scattering and improves detection accuracy, efficiency, and sensitivity.

Specific Embodiments

In order to more clearly understand the above objectives, features, and advantages of the present invention, the present invention will be further described in detail below with reference to the accompanying drawings and specific embodiments.

As shown in the technical solution, when a crowd panics due to an emergency, they usually scatter and evacuate. At this time, the change in the average velocity of pedestrians in the crowd is more drastic, the

pedestrian vector direction is more chaotic and disordered, and the crowd confusion degree evaluation index increases significantly. The threshold for the crowd confusion degree evaluation index can be preset according to actual conditions. When it is greater than the preset threshold, it is determined as an abnormal warning of crowd panic scattering.

It should be noted that performing moving target detection based on real-time video streams to obtain the motion trajectories of pedestrians in video surveillance is well-known common knowledge and existing technology in the art, and can be implemented using Yolo and Deep-sort algorithms.

Optionally, step S2 comprises: calculating the velocity v_j of each pedestrian; calculating the overall average velocity $\bar{v_t} = \Sigma_{j=1}^{N} v_j$; and obtaining the crowd velocity factor $V_t = \partial \bar{v_t} / \partial t$. In the case of crowd panic scattering, the overall average velocity of the crowd changes more drastically, and the corresponding crowd velocity factor also increases, which is positively correlated with the degree of crowd panic scattering.

Furthermore, step S5 includes $L_t = aW_t + (1 - a)V_t$, wherein a represents an adjustment parameter. By introducing this parameter, both the velocity factor and information entropy are associated with the crowd confusion degree evaluation index, comprehensively considering changes in crowd average velocity and pedestrian direction confusion. Setting the adjustment parameter to 0.5 balances their influence for most detection scenarios.